

FLSM – Fixed Length Subnet Mask

Email: **Need 2, 12, 30 and 60 IP Addresses for Departments.**

Ans: In case of FLSM,
We consider and will
allot IP Addresses on
the basis of Maximum
IP Addresses required
For Separate departments
like 60 is maximum IP in
2, 12, 30 and 60 IP

2	3 rd Floor
12	2 nd Floor
30	1 st Floor
60	Ground Floor

Now, We will use the Class C network 192.168.1.0 and do the Subnetting on the basis of 60 IP Addresses.

Step 1: Find the Host bit “n”:

$(2^n) - 2 \geq 60$ Satisfy Equation by putting

$(2^6) - 2 \geq 60$ values of N as 0 1 2 3 4 5 6 7 8

$(64) - 2 \geq 60$

$62 \geq 60$, here $n = 6$

Last 6 Bits of 32 Bits will be used as the host Bit.

STEP 2: Default Subnet Mask to Custom Subnet mask

- According to the Step 1, there are total 6 Host bits
- Host Bits Represented by 0

Default Subnet Mask: **255.255.255.0**

11111111.11111111.11111111.00000000



Moving Right to Left and turn 6 bits “0” and Remaining 1

11111111.11111111.11111111.11000000

Customized Subnet Mask: **255.255.255.192**

Count the 1s in the above bit format,

CIDR Value = /26

Step 3: Block Size = $2^n = 2^6 = 64$

Allotting the IP Address:

Ground Floor: Required = 60 IP Addresses, Wastage = 0

192.168.10.0/26 - Network IP



192.168.10.63/26 - Broadcast IP

1st Floor : Required = 30 IP Addresses, Wastage = 32

192.168.10.64/26 - Network IP



192.168.10.127/26 - Broadcast IP

2nd Floor: Required = 12 IP Addresses, Wastage = 50

192.168.10.128/26 - Network IP

192.168.10.129/26

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192.168.10.190/26

192.168.10.191/26 - Broadcast IP

Valid IP Addresses = 62

3rd Floor : Required = 2 IP Addresses, Wastage = 60

192.168.10.192/26 - Network IP

192.168.10.193/26

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192.168.10.254/26

192.168.10.255/26 - Broadcast IP

Valid IP Addresses = 62

Total Wastage:

Still We have wastage of IP Addresses in each department except of ground floor. Which Are as follows:

Ground Floor: $62 - 62 = 0$ IP

1st Floor: $62 - 30 = 32$ IP

2nd Floor: $62 - 12 = 50$ IP

4th Floor: $62 - 2 = 60$ IP

Total Wastage

$0 + 32 + 50 + 60 = 142$ IP

Required IP = 2, (62 Allotted),	3 rd Floor
Required IP = 12, (62 Allotted)	2 nd Floor
Required IP = 30, (62 Allotted)	1 st Floor
Required IP = 60, (62 Allotted)	

Ground